Assignment 3 Solution

Q.1. What is the key difference between classical set union and fuzzy set union?

(a) Classical set union considers only the elements present in both sets, while fuzzy set union considers membership values too.

(b) Classical set union involves finding the minimum of membership values for corresponding elements.

(c) Fuzzy set union disregards membership values and focuses only on elements present in both sets.

(d) None of the above

Solution: (a)

Q.2. In the context of fuzzy sets, what does the term "intersection" refer to?

(a) Evaluating the maximum of membership values for corresponding elements.

(b) Finding the common elements between two fuzzy sets.

(c) Computing the difference between two fuzzy sets.

(d) None of the above

Solution: (b)

Q.3. How is the difference of fuzzy sets computed?

(a) By taking the maximum of membership values for corresponding elements.

(b) By finding the common elements between two fuzzy sets.

(c) By summing up membership values for corresponding elements.

(d) By computing the minimum of membership values for corresponding elements.

(e) None of the above

Solution: (e)

Q.4. What property of sets is violated by fuzzy sets?

(a) Law of Contradiction

(b) Commutativity

(c) Involution

(d) None of the above

Solution: (a)

Q.5. Which property ensures that the intersection of a set with its complement results in a null set for crisp sets, but not for fuzzy sets?

(a) Idempotency

(b) Law of Contradiction

(c) Law of Excluded Middle

(d) None of the above

Solution: (b)

Q.6. What does the Law of Excluded Middle state regarding the union of a set with its complement?

(a) It always results in a null set.

(b) It always results in the set itself.

(c) It never results in the set itself.

(d) None of the above

Solution: (c)

Q.7. Which property ensures that taking the intersection of a set with itself returns the same set?

(a) Law of Contradiction

(b) Law of Excluded Middle

(c) Idempotency

(d) None of the above

Solution: (c)

Q.8. What does the involution property state regarding taking the double complement of a set?

(a) It always results in a null set.

(b) It always results in the set itself.

(c) It never results in the set itself.

(d) None of the above

Solution: (b)

Q.9. Which condition determines the outcome of A union B union C when verifying the associativity property for union in fuzzy sets?

(a) Taking the minimum of corresponding membership values

(b) Applying the maximum of corresponding membership values

(c) Finding the average of membership values

(d) None of the above

Solution: (b)

Q.10. For crisp sets A, B, and C, which property is valid, stating that the intersection of A with the union of B and C equals the union of the intersection of A and B with the intersection of A and C?

(a) Distributivity of union over intersection

(b) Distributivity of intersection over union

(c) Commutativity

(d) None of the above

Solution: (b)